Final-state $g \to gg$ contribution (Group II, Row IV): the complete analysis in coordinate space

Abstract

We redo, step by step and entirely in coordinate space, the analysis of the note *"Divergence structure of the final-state $g \to gg$ contribution"*: exact reduction, the target-rapidity (JIMWLK) and projectile-rapidity (BFKL) endpoints, the collinear (DGLAP) pole and its plus prescription, the $\beta_0\,\delta(1-\xi)$ completion and the coordinate-space running coupling, the net-colour large-distance IR divergence, and the finite remainder. The only momentum-space input is the closed form of one coordinate kernel $H^{mm'}(s)$, the coordinate image of the $\overline{\rm MS}$-regulated splitting kernel. All results agree with the note, and the endpoint coefficients agree with the ones obtained earlier in coordinate space.

1 Starting point, notation and scheme

The contribution is

$$\frac{d^3N}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{4\pi^4}\frac{1}{k^+}\int_{\xi_V}^{\xi_\Lambda}\frac{d\xi}{2\pi}\int_{z,w,w',x'} e^{-ik\cdot(w'-w)}\int_{x,y,y'}\frac{(w-z)^m(x-y)^k(w'-z)^{m'}(x'-y')^{k'}}{(w-z)^2(x-y)^2(w'-z)^2(x'-y')^2}$$
$$\times\,\delta^{(2)}\big[y-z-\xi(w-z)\big]\,\delta^{(2)}\big[y'-z-\xi(w'-z)\big]\,T_{mkm'k'}(\xi)\,\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x), \tag{1}$$

with $T_{mkm'k'} = d_\perp\xi(1-\xi)\delta_{km}\delta_{k'm'}+2(\delta_{mk'}\delta_{km'}-\delta_{km}\delta_{k'm'})+c(\xi)\delta_{mm'}\delta_{kk'}$, $c(\xi)=\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}$, $\xi_V=\frac{k^+}{\vee}$, $\xi_\Lambda=\frac{k^+}{k^++\Lambda}$.

Notation (coordinate space). The LO amplitude and its bilinears:

$$A_i^c(y)=\int_x\frac{(x-y)_i}{(x-y)^2}\big[U(y)-U(x)\big]^{ce}\rho^e(x),\qquad \mathcal{A}(y,y')\equiv A^c(y)\cdot A^c(y'),\qquad \mathcal{A}_{\hat s}(y,y')\equiv\big(\hat s\cdot A^c(y)\big)\big(\hat s\cdot A^c(y')\big), \tag{2}$$

with $\hat s=(y'-y)/|y'-y|$ and the ordering $\rho^{e'}(x')\rho^e(x)$ of Eq. (1) kept throughout. Explicitly $\mathcal{A}(y,y')=\int_{x,x'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}[U^{ce'}(y')-U^{ce'}(x')][U^{ce}(y)-U^{ce}(x)]\rho^{e'}(x')\rho^e(x)$. The LO spectrum at transverse momentum p and the prefactor are

$$\mathcal{L}(p)\equiv\int_{y,y'}e^{-ip\cdot(y'-y)}\,\mathcal{A}(y,y'),\qquad \mathcal{N}\equiv\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}=\frac{\bar C}{k^+},\qquad \hat P(\xi)=\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}+\xi(1-\xi). \tag{3}$$

Scheme (as in the note). $d=2-2\epsilon$, $d_\perp=d$; the collinear divergence is an IR divergence of the *relative* transverse momentum of the pair, so $\epsilon<0$; $\overline{\rm MS}$ measure $\mu^{2\epsilon}(e^{\gamma_E}/4\pi)^\epsilon(2\pi)^{2\epsilon}\int d^d\ell$; scaleless virtual integrals are set to zero ($1/\epsilon_{\rm UV}=1/\epsilon_{\rm IR}$). In coordinate space this means: the relative coordinate r of the two daughters (conjugate to ℓ) is the dimensionally continued variable. Rapidity divergences use the cutoffs ξ_V,ξ_Λ.

2 Exact reduction in coordinate space

Change of variables. Use the δ-functions to trade (z,w,w') for the parents (y,y') and the daughters' relative coordinate $r\equiv w-z$:

$$z=y-\xi r,\qquad w=y+(1-\xi)r,\qquad w'-z=r+s,\qquad s\equiv\frac{y'-y}{\xi}=w'-w. \tag{4}$$

The Jacobian is $|\det\partial(z,w,w')/\partial(y,y',r)|=1/\xi$ per transverse component, i.e. $1/\xi^2$ in two dimensions. The phase becomes $e^{-ik\cdot(w'-w)}=e^{-ik\cdot(y'-y)/\xi}$, the two emission kernels become $\frac{r^m}{r^2}$ and $\frac{(r+s)^{m'}}{(r+s)^2}$, and $A(y)A(y')$ does

not depend on r. Hence, exactly,

$$\frac{d^3N}{d^3k} = \frac{\bar{C}}{k^+}\int_{\xi_V}^{\xi_\Lambda} d\xi\, I(\xi), \qquad I(\xi) = \frac{1}{\xi^2}\int_{y,y'} e^{-ik\cdot(y'-y)/\xi}\, T_{mkm'k'}(\xi)\, H^{mm'}\Big(\frac{y'-y}{\xi}\Big)\, A^c_k(y) A^c_{k'}(y'), \tag{5}$$

with the *daughter kernel*

$$H^{mm'}(s) \equiv \int_r \frac{r^m}{r^2}\,\frac{(r+s)^{m'}}{(r+s)^2}. \tag{6}$$

In two dimensions H diverges logarithmically at large $|r|$: this is the collinear divergence (large daughter separation $\leftrightarrow$ small relative momentum).

Antisymmetric structure. $H^{mm'}(s) = H^{m'm}(s)$ (substitute $r \to -r-s$), so the contraction with $2(\delta_{mk'}\delta_{km'} - \delta_{km}\delta_{k'm'})$ vanishes identically, at every (ξ, y, y').

Closed form of H (the only regulated object). In the scheme above,

$$H^{mm'}(s) = -\frac{\pi}{\epsilon}\frac{\delta^{mm'}}{d} - \frac{\pi}{2}\,\delta^{mm'}\ln\frac{\mu^2 s^2 e^{2\gamma_E}}{4} - \pi\Big(\hat{s}^m\hat{s}^{m'} - \frac{1}{2}\delta^{mm'}\Big) + O(\epsilon), \tag{7}$$

using $\int d^2\ell\,[e^{i\ell\cdot s} - \theta(\mu - |\ell|)]/\ell^2 = -\pi\ln(\mu^2 s^2 e^{2\gamma_E}/4)$ and $\int d^2\ell\, e^{i\ell\cdot s}(\ell^m\ell^{m'} - \frac{1}{2}\delta^{mm'}\ell^2)/\ell^4 = -\pi(\hat{s}^m\hat{s}^{m'} - \frac{1}{2}\delta^{mm'})$ (equivalently $\mathrm{FT}[\delta - 2\hat{r}\hat{r}] = \frac{4\pi}{q^2}(2\hat{q}\hat{q} - \delta)$).

Contraction. With $d_\perp = d$ the pole contracts to exactly $\hat{P}(\xi)$; the finite trace gives $\hat{P}$ times the log; the traceless part survives only in the $d_\perp$ channel ($d_\perp \to 2$ there, no pole). With $s^2 = |y'-y|^2/\xi^2$:

$$\boxed{I(\xi) = -\pi\,\hat{P}(\xi)\,G(\xi)\;+\;I_{\mathrm{az}}(\xi),} \tag{8}$$

$$G(\xi) \equiv \frac{1}{\xi^2}\int_{y,y'} e^{-ik\cdot(y'-y)/\xi}\Big[\frac{1}{\epsilon} + \ln\frac{\mu^2|y'-y|^2 e^{2\gamma_E}}{4\xi^2}\Big]\mathcal{A}(y,y'), \tag{9}$$

$$I_{\mathrm{az}}(\xi) \equiv \pi\,\frac{1-\xi}{\xi}\int_{y,y'} e^{-ik\cdot(y'-y)/\xi}\Big[\mathcal{A}(y,y') - 2\mathcal{A}_{\hat{s}}(y,y')\Big]. \tag{10}$$

This is the note's Eq. (50), derived here without leaving coordinate space.

Remark (the ξ^2 inside the logarithm). The collinear scale is $|w'-w| = |y'-y|/\xi$, the separation of the observed gluons. Continuing z at fixed parents instead of r produces $\ln\mu^2|y'-y|^2$ without the ξ^2, because $d^dz = \xi^d d^dr$. The two choices differ by the finite term $+\pi\int d\xi\,\hat{P}(\xi)\ln\xi^2\,\mathcal{L}(k/\xi)/\xi^2$. Only the r-continuation, which is the relative-momentum $\overline{\mathrm{MS}}$ scheme, matches the $\overline{\mathrm{MS}}$ fragmentation function. Everything else agrees with the coordinate-space results obtained earlier:

- the trace $-\pi[\frac{1}{\epsilon} + \ln]$ is your $\int_z K = -\pi L_\epsilon$ (with $2/\epsilon_{\mathrm{yours}} = 1/\epsilon$);
- I_{az} is exactly your bracket term $C\frac{1-\xi}{\xi}[\mathcal{K}\cdot\mathcal{K}' - 2(\mathcal{K}\cdot\hat{r})(\mathcal{K}'\cdot\hat{r})]$.

3 Longitudinal endpoints in coordinate space

$\xi \to 1$ **(target rapidity, JIMWLK).** At $\xi = 1$, $y \to w$ and $y' \to w'$ and $s = y' - y$:

$$\mathcal{J}(k) \equiv \lim_{\xi\to 1}(1-\xi)I(\xi) = -\pi G(1) = \int_{z,w,w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\,\mathcal{A}(w,w') = -\pi\int_{w,w'} e^{-ik\cdot(w'-w)}\Big[\frac{1}{\epsilon} + \ln\frac{\mu^2|w'-w|^2 e}{4}$$

(11)

$\xi \to 0$ **(projectile rapidity, BFKL).** Since $\xi\hat{P} \to 1$, the coefficient is $-\pi G(0^+)$. Put $y' = y + \xi t$, so $d^2y' = \xi^2 d^2t$:

$$G(\xi) = \int_y\int_t e^{-ik\cdot t}\Big[\frac{1}{\epsilon} + \ln\frac{\mu^2 t^2 e^{2\gamma_E}}{4}\Big]\mathcal{A}(y, y+\xi t) \xrightarrow{\xi\to 0} \int_y \mathcal{A}(y,y)\int_t e^{-ik\cdot t}\ln t^2 = -\frac{4\pi}{k^2}\int_y \mathcal{A}(y,y), \tag{12}$$

using $\int d^2t\, e^{-ik\cdot t}\ln t^2 = -4\pi/k^2$ ($k \neq 0$; the constant and the pole multiply $\delta^{(2)}(k)$). Hence

$$\mathcal{B}(k) \equiv \lim_{\xi\to 0}\xi I(\xi) = -\pi G(0^+) = \frac{4\pi^2}{k^2}\int_z \mathcal{A}(z,z) = \int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\big[U(z) - U(x')\big]^{ce'}\Big[$$

(13)

$\mathcal{B}$ has no collinear pole. $I_{\rm az}$ contributes to neither endpoint. Near $\xi \to 0$ the traceless bracket has vanishing angular average and its transform is $\propto 1/q^2$ at $q = k/\xi$, so $\xi I_{\rm az} = O(\xi^2)$. Near $\xi \to 1$ it carries $(1-\xi)$.

Endpoint subtraction. Use the exact decomposition $\hat{P}(\xi) = \frac{1}{1-\xi} + \frac{1}{\xi} + \big(\xi(1-\xi) - 2\big)$:

$$\int_{\xi_V}^{\xi_\Lambda} d\xi\, I(\xi) = \mathcal{J} \ln \frac{1-\xi_V}{1-\xi_\Lambda} + \mathcal{B} \ln \frac{\xi_\Lambda}{\xi_V} - \pi \int_{\xi_V}^{\xi_\Lambda} d\xi \Big[\frac{G(\xi) - G(1)}{1-\xi} + \frac{G(\xi) - G(0^+)}{\xi} + \big(\xi(1-\xi) - 2\big) G(\xi) \Big] + \int_{\xi_V}^{\xi_\Lambda} d\xi\, I_{\rm az}(\xi), \tag{14}$$

with $\ln \frac{1-\xi_V}{1-\xi_\Lambda} = \ln \frac{k^+}{\Lambda} + \ln(1 - \frac{k^+}{\vee}) + O(\frac{\Lambda}{k^+})$ and $\ln \frac{\xi_\Lambda}{\xi_V} = \ln \frac{\vee}{k^+} + O(\frac{\Lambda}{k^+})$. The remaining integrals converge at both ends, so their limits can be taken to $[0,1]$ at the cost of $O(k^+/\vee, \Lambda/k^+)$. (Your earlier finite-$\vee$ form, with subtractions $\xi\mathcal{J}/(1-\xi)$ and $(1-\xi)\mathcal{B}/\xi$, is Eq. (14) rearranged. It differs by $-(1-\frac{k^+}{\vee})(\mathcal{J}+\mathcal{B})$ and the $\ln(1-\frac{k^+}{\vee})\mathcal{J}$ term, which is exact.)

4 Collinear pole, plus prescription, and the overlap

Split $G = \frac{1}{\epsilon} h + G_{\rm fin}$ with $h(\xi) \equiv \mathcal{L}(k/\xi)/\xi^2$, $h(0^+) = 0$ (since $\mathcal{L}(p) \sim p^{-4}$), $h(1) = \mathcal{L}(k)$. The pole of Eq. (14) is

$$-\frac{\pi}{\epsilon} \Big\{ \ln \frac{k^+}{\Lambda} \mathcal{L}(k) + \int_0^1 d\xi \Big[\frac{h - h(1)}{1-\xi} + \frac{h}{\xi} + \big(\xi(1-\xi) - 2\big) h \Big] \Big\} = -\frac{\pi}{\epsilon} \Big\{ \ln \frac{k^+}{\Lambda} \mathcal{L}(k) + \int_0^1 \frac{d\xi}{\xi^2}\, \hat{P}_+(\xi)\, \mathcal{L}(k/\xi) \Big\}, \tag{15}$$

where $\hat{P}_+ = \frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)$ uses the *standard* plus distribution, acting on the full test function. The two integrands agree pointwise (checked symbolically).

- The $\xi \to 0$ end of the collinear convolution converges ($h \sim \xi^2$), so the collinear and projectile-rapidity divergences do not overlap.
- The double divergence $-\frac{\pi}{\epsilon} \ln \frac{k^+}{\Lambda} \mathcal{L}(k)$ is counted once, inside $\mathcal{J}$. The plus prescription prevents counting it a second time.

In DGLAP language, with the note's normalisation ($\mathcal{N}_{\rm LO} = \alpha_s/4\pi^4$), $-\frac{\pi \bar{C}}{\epsilon k^+} \int \frac{d\xi}{\xi^2} \hat{P}_+ \mathcal{L}(k/\xi) = -\frac{1}{\epsilon} \frac{\alpha_s}{2\pi} \int \frac{d\xi}{\xi^3} [2N_c \hat{P}_+(\xi)] \frac{d^3 N_{\rm LO}}{d^3 p}\big|_{p=k/\xi}$ the real part of $P_{gg} \otimes {\rm LO}$, with ξ^{-3} the fragmentation Jacobian.

No UV divergence. In coordinate space the short-distance points $z \to w$ and $z \to w'$ ($r \to 0$, $r \to -s$) are $1/|r|$ singularities, integrable in $d = 2$. The only transverse divergences of H are at large $|r|$ (IR).

5 The $\delta(1-\xi)$ completion and the coordinate-space running coupling

$\hat{P}_+ = \frac{P_{gg}}{2N_c} - \frac{11}{12}\delta(1-\xi)$ ($n_f = 0$, $\beta_0 = \frac{11 N_c}{3}$; momentum sum rule $\int_0^1 \xi \hat{P}_+ = -\frac{11}{12}$). Add and subtract:

$$-\pi \int d\xi\, \hat{P}\, G = -\pi \int d\xi\, \frac{P_{gg}(\xi)}{2N_c}\, G(\xi) \;+\; \pi\, \frac{11}{12}\, G(1) \;-\; \pi \ln \frac{k^+}{\Lambda}\, G(1). \tag{16}$$

- *Fragmentation counterterm* at μ_F turns the first term into $-\pi \int d\xi \frac{P_{gg}}{2N_c} \frac{1}{\xi^2} \int_{y,y'} e^{-ik\cdot(y'-y)/\xi} \ln \frac{\mu_F^2 |y'-y|^2 e^{2\gamma_E}}{4\xi^2} \mathcal{A}(y,y')$.
- *Coupling counterterm* at μ_R turns the second into $+\pi \frac{11}{12} \int_{y,y'} e^{-ik\cdot(y'-y)} \ln \frac{\mu_R^2 |y'-y|^2 e^{2\gamma_E}}{4} \mathcal{A}(y,y')$, i.e. $\frac{\alpha_s \beta_0}{4\pi} \ln \frac{\mu_R^2 |y'-y|^2 e^{2\gamma_E}}{4}$ inside the LO coordinate integral. Combined with $\alpha_s(\mu_R)$ in the LO prefactor,

$$\frac{d^3 N}{d^3 k} \supset \frac{\mathcal{N}'}{k^+} \int_{y,y'} \alpha_s\Big(\frac{4e^{-2\gamma_E}}{|y'-y|^2}\Big)\, e^{-ik\cdot(y'-y)}\, \mathcal{A}(y,y'), \qquad \mathcal{N}' = \frac{1}{4\pi^4}\,: \tag{17}$$

  the coupling runs at the parent separation $|y'-y|$ *inside* the Fourier integral, not at k^2.
- The third term, $-\pi \ln \frac{k^+}{\Lambda} G(1) = \ln \frac{k^+}{\Lambda} \mathcal{J}$, carries the pole $-\frac{\pi}{\epsilon} \ln \frac{k^+}{\Lambda} \mathcal{L}(k)$. It must cancel against the $z \to w$ UV poles of the soft-gluon-scattering ($J_L U(z) J_R$-type) terms, which are not part of this row.

This term has no UV pole of its own. The add-and-subtract is the required bookkeeping once scaleless virtual integrals are set to zero: the final-state self-energy has the would-be value $\pi[\ln \frac{k^+}{\Lambda} - \frac{11}{12}]\mathcal{L}(k)(\frac{1}{\epsilon_{\rm IR}} - \frac{1}{\epsilon_{\rm UV}}) \equiv 0$.

6 Net-colour large-distance IR divergence (coordinate space)

With $U \to 1$ outside the target, at large $|y|$

$$A_i^c(y) \xrightarrow{|y|\to\infty} -\frac{y_i}{y^2}\,Q^c, \qquad Q^c = \int_x \big[\delta^{ce} - U^{ce}(x)\big]\rho^e(x), \tag{18}$$

generically nonzero, even for a colour-neutral projectile. Then $\int_{|y|<R} d^2y\,\mathcal{A}(y,y) = 2\pi Q^c Q^c \ln R + \text{const}$, so $\mathcal{B}$ diverges logarithmically at large $|z|$. In the same scheme, with $\mathcal{L}_C(k) \equiv 4\pi^2 Q^c Q^c/k^2$ (the Coulomb tail of $\mathcal{L}$),

$$\mathcal{B} = \hat{\mathcal{B}} - \frac{\pi}{\epsilon}\mathcal{L}_C(k), \qquad \mathcal{J} = \hat{\mathcal{J}}_{\rm fin} - \frac{\pi}{\epsilon}\mathcal{L}(k) - \frac{\pi}{\epsilon}\mathcal{L}_C(k), \qquad I = \hat{I} - \frac{\pi}{\epsilon}\frac{\hat{P}}{\xi^2}\mathcal{L}(k/\xi) - \frac{\pi}{\epsilon}\hat{P}\,\mathcal{L}_C(k), \tag{19}$$

(the large-$|y|, |y'|$ region of the coordinate integrals), and, using $\int_0^1[\hat{P} - \frac{1}{1-\xi} - \frac{1}{\xi}] = \int_0^1(\xi(1-\xi)-2) = -\frac{11}{6}$,

$$\frac{d^3N}{d^3k} \supset -\frac{\pi\bar{C}}{\epsilon\,k^+}\,\mathcal{L}_C(k)\Big[\ln\frac{k^+}{\Lambda} + \ln\frac{\vee}{k^+} - \frac{11}{6}\Big]. \tag{20}$$

With a large-distance cutoff, $-1/\epsilon \leftrightarrow \ln R^2$ up to a constant. This pole cannot cancel within this row.

7 Result

Compact form (exact up to $O(\Lambda/k^+, k^+/\vee)$, all transverse integrals in coordinate space):

$$\boxed{\frac{d^3N}{d^3k} = \mathcal{N}\Big\{\ln\frac{k^+}{\Lambda}\,\mathcal{J}(k) + \ln\frac{\vee}{k^+}\,\mathcal{B}(k) - \pi\int_0^1 d\xi\Big[\frac{G(\xi)-G(1)}{1-\xi} + \frac{G(\xi)-G(0^+)}{\xi} + \big(\xi(1-\xi)-2\big)G(\xi)\Big] + \int_0^1 d\xi\, I_{\rm az}(\xi)\Big\}} \tag{21}$$

with $\mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}$, and G, $I_{\rm az}$, $\mathcal{J} = -\pi G(1)$, $\mathcal{B} = -\pi G(0^+)$ from Eqs. (9), (10), (11), (13).

All poles explicit (coordinate version of the note's Eq. (65)):

$$\frac{d^3N}{d^3k} = \mathcal{N}\Big\{\underbrace{-\frac{\pi}{\epsilon}\int_0^1\frac{d\xi}{\xi^2}\Big[\hat{P}_+ + \frac{11}{12}\delta(1-\xi)\Big]\mathcal{L}(k/\xi)}_{\text{full } P_{gg}\text{: } \overline{\rm MS} \text{ fragmentation c.t.}} \underbrace{+\frac{\pi}{\epsilon}\frac{11}{12}\mathcal{L}(k)}_{\text{cancels } \alpha_s \text{ c.t.}} \underbrace{-\frac{\pi}{\epsilon}\ln\frac{k^+}{\Lambda}\mathcal{L}(k)}_{\text{UV of target evolution}} \underbrace{-\frac{\pi}{\epsilon}\mathcal{L}_C(k)\Big[\ln\frac{k^+}{\Lambda} + \ln\frac{\vee}{k^+} - \frac{11}{6}\Big]}_{\text{net-colour IR}}$$
$$+ \ln\frac{k^+}{\Lambda}\,\hat{\mathcal{J}}_{\rm fin}(k) + \ln\frac{\vee}{k^+}\,\hat{\mathcal{B}}(k) + \hat{F}(k;\mu)\Big\}, \tag{22}$$

with the coordinate-space finite pieces

$$\mathcal{J}_{\rm fin}(k) = -\pi\int_{w,w'} e^{-ik\cdot(w'-w)}\ln\frac{\mu^2|w'-w|^2e^{2\gamma_E}}{4}\,\mathcal{A}(w,w'), \qquad \mathcal{B}(k) = \frac{4\pi^2}{k^2}\int_z \mathcal{A}(z,z), \tag{23}$$

$$F(k;\mu) = \int_0^1 d\xi\Big[I_{\rm fin}(\xi) - \frac{\mathcal{J}_{\rm fin}(k)}{1-\xi} - \frac{\mathcal{B}(k)}{\xi}\Big], \tag{24}$$

$$I_{\rm fin}(\xi) = -\frac{\pi}{\xi^2}\int_{y,y'} e^{-ik\cdot(y'-y)/\xi}\Big\{\hat{P}(\xi)\ln\frac{\mu^2|y'-y|^2e^{2\gamma_E}}{4\xi^2}\,\mathcal{A}(y,y') + 2\xi(1-\xi)\Big[\mathcal{A}_{\hat{\rm s}}(y,y') - \frac{1}{2}\mathcal{A}(y,y')\Big]\Big\}, \tag{25}$$

and hats denoting removal of the net-colour (Coulomb-tail) pieces. After the $\overline{\rm MS}$ counterterms, the β_0 share becomes the coordinate-space running coupling of Eq. (17).

8 Checks

- Change of variables: Jacobian $1/\xi^2$ and $w'-w = (y'-y)/\xi$ (symbolic); $H^{mm'} = H^{m'm}$ kills the antisymmetric structure exactly.
- $\hat{P} - \frac{1}{1-\xi} - \frac{1}{\xi} = \xi(1-\xi) - 2$, $\int_0^1(\xi(1-\xi)-2) = -\frac{11}{6}$, and the subtracted collinear integrand equals the standard-plus integrand pointwise (symbolic); $\int_0^1 \xi\hat{P}_+ = -\frac{11}{12}$.
- $-\pi G(1) = \mathcal{J}$ is the dimensionally regularised $\int_z K_{ww'z}$; $-\pi G(0^+) = \mathcal{B}$ follows from $\mathrm{FT}[\ln t^2] = -4\pi/k^2$. These are the JIMWLK and BFKL coefficients obtained earlier, with the same signs and prefactors.

- I_{az} reproduces the earlier finite bracket term. Its momentum-space image $(4\pi/q^2)(2\hat{q}\hat{q} - \delta)$ was verified distributionally to 8 digits.
- $I(\xi)$ of Eq. (8) is the note's Eq. (50). The note checked it against the momentum-space disc-subtracted integrand to 9×10^{-7} and 3×10^{-8}.